

A Search for Extended Circumgalactic Dust

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1 Motivation

Most of the universe's atoms are hidden from plain sight: stars account for only 10% of baryons, while the remaining 90% reside in two gaseous phases that are extremely difficult to observe because of their low densities: the circumgalactic medium (CGM), filling the dark matter halos hosting galaxies, and the more extended intergalactic medium (IGM). Despite being the focus of intense theoretical and observational research, the CGM and IGM remain poorly understood. This matters because abundance and spatial extent of dust in the Universe is a key component of galaxy formation models, comprises a spatially varying foreground screen that has practical implications for supernova cosmology, and fundamentally limits the homogeneity of samples selected for wide-field surveys.

While attempting to measure magnification of 85,000 background quasars by 20 million normal galaxies in SDSS images, [4] found a smooth extinction profile extending beyond 1 Mpc. A similar measurement is performed by [5], using as their background sources SDSS Luminous Red Galaxies (LRGs), whose remarkably small dispersion in optical colors allows for high accuracy with a smaller sample size. Together, these measurements imply that more than half of the dust in the universe lies outside galactic disks. This result is highly unexpected: the presence of dust well into the two-halo regime of the matter correlation function defies the predictions of hydrodynamic simulations of dust production and survival, under-predicting the observed measurements of [4, 5] by an order of magnitude [3, 7].

Beyond these measurements, investigators have searched for dust-induced extinction of background sources near a galaxy (Zaritsky 1994), or in stacks of clusters (Chelouche et al. 2007, Bovy et al. 2008). A direct detection of circumgalactic dust (CGD) at the ≥ 200 kpc radii probed by the work of [4, 5] would allow for a meaningful comparison to the simulations of cosmological dust abundance such as the ones by [3, 7].

The abundance and spatial extent of dust in the universe is a key component of galaxy formation models. The presence of foreground dust fundamentally limits the homogeneity

of samples in wide-field surveys, and has practical implications for cosmology supernovae and cluster weak lensing. Accordingly, errors in foreground dust models will have profound knock-on effects. XXX which ones? XXX

In this context, [4] made an unusual discovery while attempting to measure the magnification of 85,000 background quasars by 20 million galaxies in SDSS images. They found a smooth reddening profile detectable out to 10 Mpc from the foreground galaxies; they ascribe this reddening to a circumgalactic dust halo at cosmological scales. A handful of other studies (e.g, [5]) also find evidence for excess reddening by foreground galaxies. These results implies that more than half of the dust in the universe lies outside galactic disks, and are in strong tension with the predictions of hydrodynamic simulations of dust processing and survival.

If this excess dust is real, it has serious implications for all cluster cosmology surveys. All weak lensing analysis depends on accurate galaxy colors to obtain distance estimates, both for creating background galaxy samples and for fitting masses to the galaxy shear signal. A halo of dust at cosmological scales would skew the models used to turn galaxy colors into distances, and may ultimately bias cosmological parameter constraints. An extended halo would also indicate the need to revisit assumptions about how dust enters the CGM, and how long it can survive.

2 Methods

A screen of dust will Rayleigh scatter incoming light, resulting in an attenuation or extinction and changing its spectrum by reddening it. The total extinction A_λ quantifies wavelength-depended dust absorption; reddening is usually parametrized by the selective extinction red versus blue light: $E(B - V) = (B - V)_{\text{meas}} - (B - V)_0$. The steepness of A_λ as a function of wavelength is usually parametrized by

$$R_V = A_V/E(B - V) \tag{1}$$

For interstellar Milky Way dust, $R_V \sim 3.1$; dense molecular clouds have $R_V \sim 5$.

These statements are generally true for dust-induced reddening and extinction. However, we are looking for a faint, extended signal, buried within the noise and color scatter of our background sample. Magnitude differences like $E(B - V)$ are instructive, but somewhat simplistic given the subtlety of the effect. Here, we use maximum likelihood parameter estimation to find a linear combination of wavelengths that represents the best chance of finding the extinction. This optimal estimator is applied to a sample of background galaxies that have been binned in redshift space to control for cosmological (not dust-induced) reddening. Finally, the resulting dust extinction measure is correlated with the presence of foreground galaxies to search . We describe our approach in greater detail

below, noting that for the sake of readability all vector or matrix quantities appear in boldface.

2.1 A Maximum Likelihood Estimator for Dust Extinction

For a uniform prior probability distribution of a given parameter, the log-likelihood of $\mathbf{D}$ at a given value is ρ :

$$\log L = [\mathbf{D} - \mathbf{M}(\rho)]^T \frac{\mathbf{C}^{-1}}{2} [\mathbf{D} - \mathbf{M}(\rho)] \quad (2)$$

For galaxies, the data vector $\mathbf{D}$ is an array of magnitudes, e.g.,

$$\mathbf{D} = \{u, g, r, i, z, W1, W2, \dots\} \quad (3)$$

while the covariance of the data $\mathbf{C}$ encapsulates both the magnitude measurement uncertainties and cosmic variance. The term $\mathbf{M}(\rho)$ is the best-fit model describing the data $\mathbf{D}$; ρ is the unknown parameter for which Equation 2 is solved – here, a unit A_V . Taylor expansion of $\mathbf{M}(\rho)$ yields a simple linear approximation for the effect of ρ :

$$\mathbf{M} = \mathbf{M}_0 + \rho \frac{\partial \mathbf{M}}{\partial \rho} \quad (4)$$

In this expansion, $\mathbf{M}_0$ are the intrinsic colors of galaxies in the background galaxy sample, without any reddening from dust. The $\rho \frac{\partial \mathbf{M}}{\partial \rho}$ term is the response of galaxies to a unit dust; $\frac{\partial \mathbf{M}}{\partial \rho}$ is a fiducial (theoretical) extinction curve for dust. To replicate the work of [4], we use the extinction model of [1] for $R_V = 3.1$ Milky Way-type dust.

The optimal estimator $\hat{\rho}$ for the dust extinction ρ , such that ρ maximizes the likelihood in Equation 2, is given as:

$$\hat{\rho} = \left[\frac{\frac{\partial \mathbf{M}}{\partial \rho}^T \mathbf{C}^{-1} (\mathbf{D} - \mathbf{M}_0)}{\frac{\partial \mathbf{M}}{\partial \rho}^T \mathbf{C}^{-1} \frac{\partial \mathbf{M}}{\partial \rho}} \right] \quad (5)$$

where by the Cramér-Rao bound, the uncertainty on $\hat{\rho}$ is given by:

$$\sigma_{\hat{\rho}}^2 = \frac{\partial \mathbf{M}}{\partial \rho}^T \mathbf{C}^{-1} \frac{\partial \mathbf{M}}{\partial \rho} \quad (6)$$

The argument leading to Equations 5 and 6 can also be presented as follows. Each galaxy in the (background) sample has a set of colors

$$\mathbf{D} = \mathbf{M}_0 + A_V d\mathbf{M} + \text{noise} \quad (7)$$

The noise term is independent of A_V , and behaves as a multivariate Gaussian with a covariance $\mathbf{C}$. In the absence of dust, $\mathbf{D}$ minus the mean color of the sample M_0 is simply a set of values points drawn from the distribution described by $\mathbf{C}$. The expression

$$\frac{\mathbf{C}^{-1}}{2} [\mathbf{D} - \mathbf{M}(\rho)]^T \quad (8)$$

parametrizes how significant the deviation in $\mathbf{D}$ is from M_0 . We may then ask how much of the observed deviation is in the direction we would expect from dust by projecting Equation 8 into the theoretical dust curve $\partial\mathbf{M}/\partial\rho$:

$$(\mathbf{D} - M_0) \mathbf{C}^{-1} \frac{\partial\mathbf{M}}{\partial\rho} \quad (9)$$

Rescaling (calibrating) this deviation in background galaxy color from the total dust column, relative to the reddening from a “unit dust” column, yields the final form of the estimator in Equations 5

$$\hat{\rho} = \left[\frac{\frac{\partial\mathbf{M}}{\partial\rho} \mathbf{C}^{-1} (\mathbf{D} - M_0)}{\frac{\partial\mathbf{M}}{\partial\rho} \mathbf{C}^{-1} \frac{\partial\mathbf{M}}{\partial\rho}} \right] \quad (10)$$

The factor

$$\frac{\partial\mathbf{M}/\partial\rho}{\frac{\partial\mathbf{M}}{\partial\rho} \mathbf{C}^{-1} \frac{\partial\mathbf{M}}{\partial\rho}} \quad (11)$$

is then a normalized reddening vector per “unit dust”

Equations 4 to 6 implicitly assume that the theoretical extinction curve $\partial\mathbf{M}/\partial\rho$ accurately describes the CGD. For consistency with [menard2010measure], we take a fiducial value of $\partial\mathbf{M}/\partial\rho$ that corresponds to $R_V = 3.1$ dust. However, this need not be the case – circumgalactic dust may have a different reddening vector. To test this idea, we apply the Gram-Schmidt orthogonalization procedure to the fiducial $\partial\mathbf{M}/\partial\rho$. The resulting set of orthonormal vectors span the “extinction basis,” into which the true CGD reddening vector is projected.

For observations made in the *griz* filter set, and taking the model of [1] for $R_V = 3.1$ dust, the fiducial $\partial\mathbf{M}/\partial\rho$ extinction vector is

$$\begin{aligned} \mathbf{v}_{\text{fid}} &= [1.12224688, 0.82747095, 0.62680647, 0.47880753] \\ &\equiv A_V^{\text{fid}} \end{aligned}$$

The orthonormal basis vector $\mathbf{v}_1^{\text{ON}}$ is constructed as

$$\mathbf{v}_1 = [1, 0, 0, 0] \quad (12)$$

$$\mathbf{v}'_1 = \mathbf{v}_1 - \langle \mathbf{v}_1 | \mathbf{v}_{\text{fid}} \rangle * \mathbf{v}_{\text{fid}} \quad (13)$$

$$\mathbf{v}_1^{\text{ON}} = \frac{\mathbf{v}'_1}{\|\mathbf{v}'_1\|} \quad (14)$$

The procedure is then repeated for $\mathbf{v}_2 = [0, 1, 0, 0]$ and $\mathbf{v}_2 = [0, 0, 1, 0]$, taking care to subtract the projection of the previous vectors onto each new vector. The true CGD extinction can then be reconstructed from the individual magnitudes through a simple χ^2 minimization, i.e.,

$$\chi^2 = \sum \frac{(k_{\text{fid}} - \alpha_i k_i)^2}{(\sigma_{\text{fid}}^2 + \alpha_i^2 \sigma_i^2)}, i = 1, 2, \dots \quad (15)$$

where k is the extinction signal measured along a given basis vector: either the fiducial $\partial M / \partial \rho$ (k_{fid}) or one of the orthogonal components ($k_1, k_2, \dots$) and σ is the uncertainty of the extinction measurement. The true CGD extinction is thus

$$k^{\text{true}} = k^{\text{fid}} + \alpha_1 k_1 + \alpha_2 k_2 + \alpha_3 k_3 + \dots \quad (16)$$

This orthogonalization procedure is analogous to a principal component analysis, where each basis vector captures some of the variance of the reddened galaxy colors. Positive or negative value in the individual components in ?? mean that the “principal component” $\mathbf{v}_{\text{fid}}$ either under or overfits the actual trend at that radius.

2.2 Galaxy-Reddening Correlation Function

With the dust extinction estimator above in 2.1, we may then correlate the magnitude of dust-induced extinction with the presence of a foreground galaxy. We do this with the **TreeCorr**¹ package [2], specifically the galaxy-kappa cross-correlation. Though intended to correlate galaxy counts with shear signal (i.e., κ) the code is flexible enough to accommodate any scalar quantity. Here, we correlate foreground galaxy counts with the extinction of background galaxies as determined with the estimator in Equation 5 for the fiducial dust model and its orthogonal components derived in Equation 14. A variety of other two-point correlations, such as foreground star/background galaxy extinction, are performed as null tests. Foreground and background galaxy-galaxy two-point correlation functions are also computed. All correlations are taken in twenty impact parameter bins, from 0.2 to 200 arcminutes (or 200 kiloparsecs and 10 megaparsecs in the average foreground galaxy rest frame). This range of impact parameters matches the physical scales probed in [4].

3 Datasets

- redMaGiC as background (discuss algorithm, add some plots of color variance with MLE or color-color plots).

¹<https://github.com/rmjjarvis/TreeCorr>

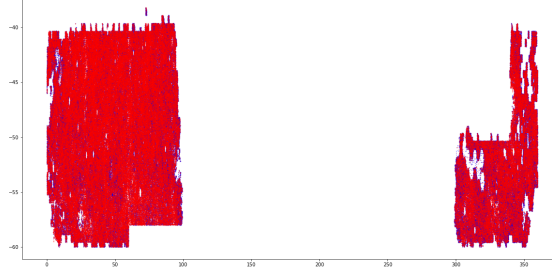


Figure 1: Example figure showing overlap of SuperCOSMOS foreground galaxies (blue points) with redMaGiC background galaxies (red points). Exact area overlap was established with the HEALPy package.

- The catalog covers the area $290^\circ \lesssim \alpha \lesssim 110^\circ$ and $-60^\circ \lesssim \delta \lesssim -40^\circ$ taken from DES Y1 survey data.
- Shoot, need to describe that redshifts come from the rmz catalog, and the photometry comes from whatever the gold MOF catalog is – ref also needed!
- OK, but first, do describe algorithm.
 - how does the redMaGiC algorithm actually work. Also, how good is the redshift fitting algorithm? This will be relevant for when we subdivide the background galaxy catalog into redshift slices, and also minimizes contamination with the foreground galaxy sample.
 - Give redshift range, filter set, etc.
 - Do these LRGs intrinsically appear in clusters? Not a lot of those in the field, “dust” saying – hopefully doesn’t bias our results
- For MOF catalog, note that magnitudes are dereddened for galactic extinction and are (I think?) based on `MAG_AUTO` of SExtractor. I appear to have lost the main catalog, but will have to dig up a reference from somewhere else.

Our background catalog for extinction measurements consists of 599,554 luminous red galaxies drawn from the Dark Energy Survey (DES, citation needed?) Year 1 data release. Galaxies are selected with the red sequence Matched-filter Galaxy Catalog (redMaGiC) algorithm [6]), which fits every galaxy with a red sequence template (itself derived from the redMaPPer red-sequence cluster finder [rykoff2014red]), computes the resulting photometric redshift and luminosity, and retains only those galaxies for which the galaxy luminosity is high and the fit is good ($\chi^2 < \chi_{\max}^2$). Rather than a hard cutoff, the redMaGiC selection threshold $\chi_{\max}^2$ is allowed to float with redshift to assure a constant co-moving galaxy density. Photometric redshifts of redMaGiC galaxies have less than 1% bias rela-

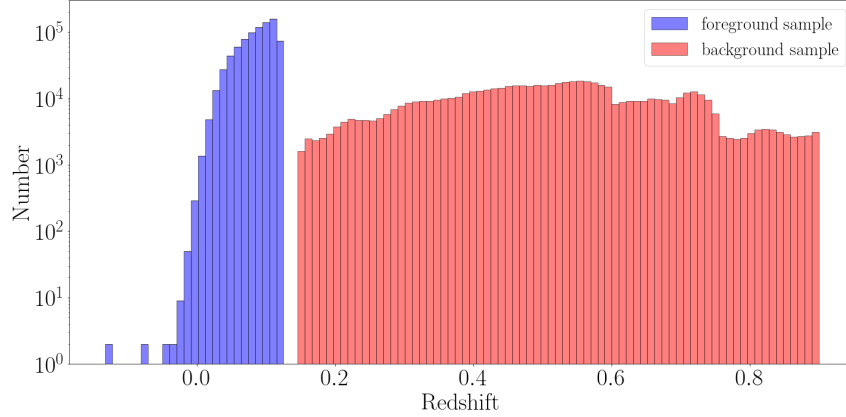


Figure 2: Redshift distribution of the SuperCOSMOS foreground and redMaGiC background galaxy catalogs

tive to spectroscopic redshifts over the entire range of the catalog ($0.15 < z < 0.85$). The distribution of redshifts is shown in Figure 3.

redMaGiC contains redshifts and errors, but no photometry. We thus match against the DES Year 1 'y1a1-gold-mof-badregion.fits' Red sequence galaxies have the advantage of a relatively tight dispersion in color space, as shown in Figure ??.

The redMaGiC data includes the standard Dark Energy Survey *griz* bands which have been dereddened of galactic extinction; due to the generally poor throughput of the Dark Energy Camera in *u*, the catalog does not include data in that filter.

The DES masking makes the survey area of redMaGiC galaxies highly inhomogeneous. A mismatch between the foreground and background areas would mean correlating over empty space and diluting the extinction signal. To ensure that the spatial distribution of the foreground galaxies (and randoms) is exactly the same as the redMaGiC galaxies, we used HEALPy² software. HEALPy is a Python implementation of the Hierarchical Equal Area isoLatitude Pixelization (HEALPix) scheme to manage discrete data on the sphere. HEALPy is used to pixelate the foreground catalogs and cross-match with the HEALPix map that is supplied with the redMaGiC galaxy catalog. The final foreground catalogs contain only galaxies for which the corresponding redMaGiC HEALPix pixel is not empty. An example with the SuperCOSMOS galaxy catalog is shown in Figure ??.

- Foregrounds:
 - SuperCOSMOS

²<https://healpy.readthedocs.io/en/latest/>

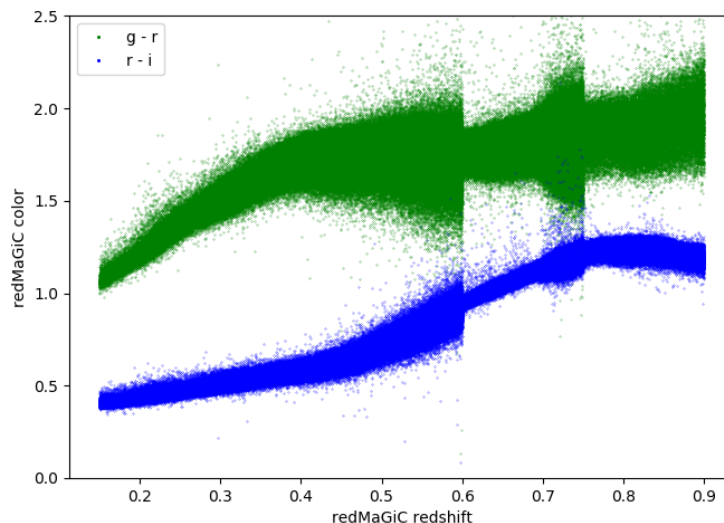


Figure 3: Color of redMaGiC galaxies as a function of redshift. At $z < 0.25$, $g - r$ color (green points) is a fairly good indicator of the red sequence (~ 0.05 , though $r - i$ color is also

- * contains B, R, W1-4 and very good photo-zs
- GALEX
 - * Take a NUV magnitude cut at 21.0 I thought? Centers photo- z distribution at 0.15. Magnitude cut of NUV ≤ 21.0 centers the distribution at $z_{\text{phot}} = 0.1$ though it decreases the sample size sharply.
- Also SVM for DES foreground sample – brief discussion as that didn't work.
 - give some characteristics
 - perhaps move discussion of algorithm to 2
 - * ref needed

The Galaxy Evolution Explorer (GALEX) was a NASA mission that imaged 77% of the sky in at least one of its near-UV (NUV, 232 nm) and far-UV (FUV, 154 nm) bands.

4 Results

- Plot up all the results I do have... with dust vector, "optimal" vector for all samples

- Don't forget to discuss the A_V Menard fit, and how I rescale it for different sample redshifts
- MB show the "dust mass plot" but maybe also skip because it's not really a dust mass
- Star-bkg correlation is a good null test to show, as is very interesting red magic autocorrelations

The variance of ξ , only including the shot noise propagated into the final correlation. This does not include sample variance, so it is always an underestimate of the actual variance. Figure ?? shows that the real reddening vector lies principally in the $v_{\text{dust}}-v_0$ plane

5 Conclusions

- Start discussing consequences: here, compare to literature
- It might be cool to compare to MgII absorption plot, since that also shows evidence for extinction
 - Except that maybe that's controversial for actual MgII folks? Apparently they expect this to be anti-correlated
- For the literature figures, consider placing in appendix?

This work is part of a larger project to detect and characterize extended halos of circumgalactic dust. The next phase will be studying

this measurement constrains the average radial dust column density (and thus dust mass). We will also search for submillimeter emission from the extended dust halos of a statistical sample of galaxies. Detection of this extended dust emission, separate from the ISM of the galaxies themselves, would be, to the best of our knowledge, the first measurement of its kind.

References

- [1] Edward L Fitzpatrick. In: *Publications of the Astronomical Society of the Pacific* 111.755 (1999), p. 63.
- [2] Mike Jarvis, Gary Bernstein, and Bhuvnesh Jain. In: *Monthly Notices of the Royal Astronomical Society* 352.1 (2004), pp. 338–352.
- [3] Ryan McKinnon, Paul Torrey, and Mark Vogelsberger. In: *Monthly Notices of the Royal Astronomical Society* 457.4 (2016), pp. 3775–3800.
- [4] Brice Ménard et al. In: *Monthly Notices of the Royal Astronomical Society* 405.2 (2010), pp. 1025–1039.

- [5] JEG Peek, Brice Ménard, and Lia Corrales. In: *The Astrophysical Journal* 813.1 (2015), p. 7.
- [6] Eduardo Rozo et al. In: *Monthly Notices of the Royal Astronomical Society* 461.2 (2016), pp. 1431–1450.
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